Pranavi Paudel

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Education

Mississippi State University
Bachelor of Science in Mechanical and Biomedical Engineering — May 2027
Cumulative GPA: 4.0/4.0, President's List 2023, 2024, 2025

Skills

- CAD, CAM & Design: SolidWorks, Inventor, AutoCad, Fusion 360, GD&T
- Simulation & Analysis: CFD, FEA, ANSYS, Minitab, Excel
- Manufacturing: Additive CNC, Additive Manufacturing, 3D Printing
- Programming & Technical Computing: Python, MATLAB, Mathematica
- Electronics & Mechatronics: Electronics, Mechatronics, PCB / EDA Design, LabVIEW

Certifications

- CSWP (SOLIDWORKS CAD Design Professional), CSWA (SOLIDWORKS CAD Design Associate), CSWA Additive Manufacturing, SolidProfessor GD&T, OSHA 10
- ACE CNC Machining Training Completion Certificate (Fusion 360)

Experience

Design and Manufacturing Engineer Intern — ***June 2026- August 2026***
Parker Hannifin Fluid Control Division — ***Madison, MS***

- Contributed to Kaizen projects generating approximately $510K in total savings by optimizing valve assembly flow, reducing waste, integrating camera-based verification, and making operations more sequential and reliable.
- Improved production efficiency by redesigning a cell that reduced cell footprint from 240 sq. ft. to 108 sq. ft., improved organization, and increased production output from 13 to 27.2 PPHL through layout optimization, time studies, 5S and yamazumi analysis.
- Developed 20+ standard work sheets, updated routers, and reduced changeover time from 100 seconds to 10 seconds by replacing a manual outdated test stand with a newly programmed system for pressure decay, mass flow, and functional valve testing.
- Designed and fabricated 3D-printed fixtures and tooling using Autodesk Inventor, SolidWorks, and Bambu Lab, while supporting PLC/HMI automation projects for operator guidance and sequential manufacturing controls.

Engineering Assistant — ***January 2026- Present***
Center for Advanced Vehicular Systems (CAVS), Mississippi State University — ***Starkville, MS***

- Designing and optimizing a ventricular shunt catheter in SolidWorks, reducing simulated flow stagnation zones by ~33% via refinement of lumen geometry, inlet hole configuration, and semi-open tip design, while accelerating design iteration cycles by ~30%.
- Performing CFD simulations in SolidWorks Flow Simulation and ANSYS Fluent under physiologically relevant flow rates (laminar regime, low Reynolds number), improving velocity distribution uniformity by ~18% and minimizing recirculation zones linked to obstruction risk.
- Applying FEA in Abaqus to evaluate structural response under physiological pressure loading, reducing peak stress concentrations by ~20% while maintaining geometric feasibility for manufacturing.

Projects

Servo-Actuated Humanoid Hand for Motion Replication

- Designed, modeled, and fabricated a multi-DOF humanoid hand using SolidWorks and 3D printing using Bambu Lab, developing finger linkages and joint mechanisms and iterating prototypes to improve fit, strength, joint mobility, and functional gripping.
- Integrated microcontroller-based control (Arduino UNO), servo motors, and sensor inputs to enable synchronized multi-joint motion with improved positional accuracy and system responsiveness (~20%) compared to initial prototype performance.
- Built and optimized the hardware and signal processing pipeline, including calibration, motion mapping, and real-time visualization, resulting in a functional prototype capable of consistent joint movement tracking and replication.

DrainWatch: Sensor-Integrated Jackson-Pratt Drain Support System

- Designed and developed a mechatronic wearable system to stabilize Jackson-Pratt drains and prevent tubing-induced displacement, improving user comfort and reducing mechanical strain during daily movement.
- Integrated a load cell-based sensing system with microcontroller control (Arduino Uno) to estimate reservoir fullness in real time without modifying the medical device, enabling non-invasive monitoring with improved measurement consistency (~20%).
- Modeled, prototyped, and iterated belt, holster, and sensor integration assemblies in SolidWorks, optimizing load transfer, fit, and manufacturability through continuous design-validation cycles, achieving ~31% improvement in overall system performance over the initial prototype, with ongoing iterative enhancements to durability, usability, and reliability.

Advanced Manufacturing Society (AMS)

- Trained and got hand-on in design-for-manufacturing workshops, coaching CAD and CAM fundamentals, certification-style modeling, and emphasizing proper datums, tolerancing, and feature design for real-world production.
- Trained on prototyping and CNC machining workflows, including drilling, milling, and lathe operations, as well as 2- axis and 3-axis CNC processes, guiding teams from CAD to finished parts with strong focus on DFM (Design for Manufacturing) and DFA (Design for Assembly).

Leadership Experience

- Tutoring in Engineering Mechanics, Mechanics of Materials, Thermodynamics, and Fluid Mechanics under Bagley College of Engineering.
- Co-authored two research papers on assistive robotic systems for mobility and locomotion, conducting analysis of 50+ systems to identify key design principles, biomechanical considerations, and optimization strategies for real-world applications.
- Completed a multi-semester leadership and service program, leading fundraising initiatives of $2000 for Boys and Girls Club being a MLP Scholar.